Miray Hirabayashi

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EDUCATION

Iowa State University **Ames, Iowa**
Bachelor of Science, Major in Software Engineering, Minor in Cybersecurity *Expected Graduation: May 2027*

- **Cumulative GPA:** 3.22

WORK EXPERIENCE

Iowa State University Department of Electrical and Computer Engineering **Ames, Iowa**
Undergraduate Teaching Assistant *Sep 2026 – Present*

- Support **20+** students with **C** programming and **Linux**-based lab exercises, troubleshooting code and explaining operating system concepts including processes, threads, scheduling, and memory management.
- Grade assignments and lab reports, reviewing students' **C** programs and **Makefiles** while evaluating their understanding of operating system concepts and providing constructive feedback.

Iowa State University Language Studies Resource Center **Ames, Iowa**
Student Assistant *Dec 2024 – Present*

- Troubleshoot hardware and software issues for students and faculty, supporting classroom technology, displays, projectors, printing/scanning systems, and related equipment.

Iowa Department of Transportation **Ames, Iowa**
Administrative Intern *May 2025 – Sep 2026*

- Developed a **Python** CLI application to automate validation and reconciliation between Workday and Microsoft Access datasets, processing **11,000+** records and eliminating manual verification.

PROJECTS

NextClose – Stock Movement Predictor | Python, FastAPI, scikit-learn, Docker, GCP (Cloud Run), Next.js, TypeScript, React, Tailwind CSS, PostgreSQL

- Built a full-stack ML web application using a Random Forest classifier trained on **35+** years of market data, with a **FastAPI** backend and **Next.js** frontend.
- Containerized and deployed the ML backend to **Google Cloud Run** using **Docker**, with **PostgreSQL** caching to reduce repeat prediction latency from **~50 seconds to near-instant** retrieval.

GlobeBase – Interactive 3D Travel Tracker | Next.js, React, TypeScript, Prisma, PostgreSQL, Three.js, Tailwind CSS

- Built a full-stack interactive 3D travel tracker rendering **250+** geographic entities with **Next.js**, **React**, **TypeScript**, and **Three.js**.
- Developed REST APIs with **Zod** validation and **Prisma/PostgreSQL** persistence, enabling users to save and synchronize travel data across devices.

WorkSync – Android App | Java, MySQL, Android Studio, Gradle

- Developed an Android employee and project management application in **Java**, integrating REST APIs, WebSockets, and **MySQL** for task and workflow management.
- Collaborated in an Agile team using **Git/GitLab**, CI/CD pipelines, feature branches, merge requests, and unit testing.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, PHP, SQL, Swift, HTML, CSS
- **Frameworks & Libraries:** React, Angular, Next.js, Node.js, Express, Flask, Prisma, Tailwind CSS, scikit-learn, FastAPI
- **Cloud & DevOps:** Google Cloud Platform (GCP), Docker, Supabase, Vercel
- **Developer Tools:** Git (GitHub, GitLab), Postman, Figma, Xcode, Android Studio
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Operating Systems & Environments:** Linux, macOS, Windows